

# Read quality and genome alignment after KRAS suppression

## WEEK 2 PROGRESS REPORT

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### Scope and computational methods

This report documents read-quality assessment and full genome alignment of 17 paired-end RNA sequencing runs from the pancreatic ductal adenocarcinoma dataset. The objective was to assess sequence quality, select strand settings and summarize alignment performance.

Raw files were organized into 17 R1/R2 pairs. Truncated imports were replaced with recovered files before analysis. FastQC 0.12.1 assessed each of the 34 files for base quality, sequence composition, duplication and adapters. MultiQC 1.35 combined 32 Galaxy reports and two verified local reports to compare results without counting duplicate reports.

HISAT2 2.2.3 aligned paired reads to the human hg38 reference. Full runs used unstranded mode, Phred+33 quality scores, no read limit, no end trimming and otherwise default alignment settings. Machine-readable summaries and aligned BAM files were retained. Overall alignment uses individual reads as its denominator; concordant unique alignment uses input pairs. Appendix A specifies the calculations.

### Read quality and strand interpretation

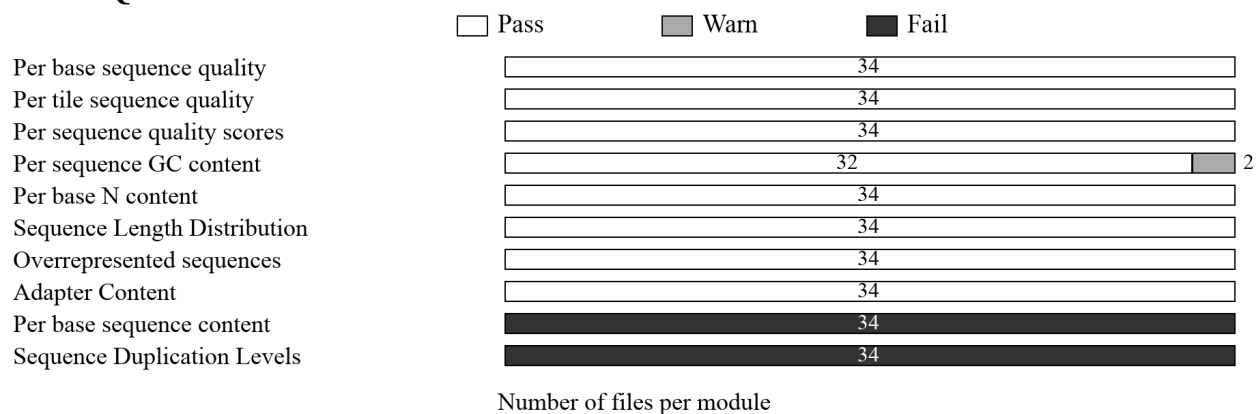
All 34 files contained 150-base reads. Base-quality and adapter checks passed in every file. Sequence composition and duplication modules flagged all files; duplication ranged from 57.0% to 67.3%. Unequal transcript abundance can contribute to duplication in RNA sequencing, so duplicate reads were not automatically removed. Both SRR24828471 mates received GC-distribution warnings; this alone does not establish contamination. Cutadapt was not applied because quality and adapter checks did not indicate a trimming requirement. Figure 1A and Appendix B summarize the checks.

Unstranded, forward-stranded (FR) and reverse-stranded (RF) pilots used the first one million pairs from Control M1 and KRAS P1. Overall alignment was 96.63% and 96.40%, respectively, under all three settings. Concordant unique alignment was 86.40% versus 86.44% (FR/RF) for Control M1 and 87.43% throughout for KRAS P1. These comparisons did not resolve strand specificity.

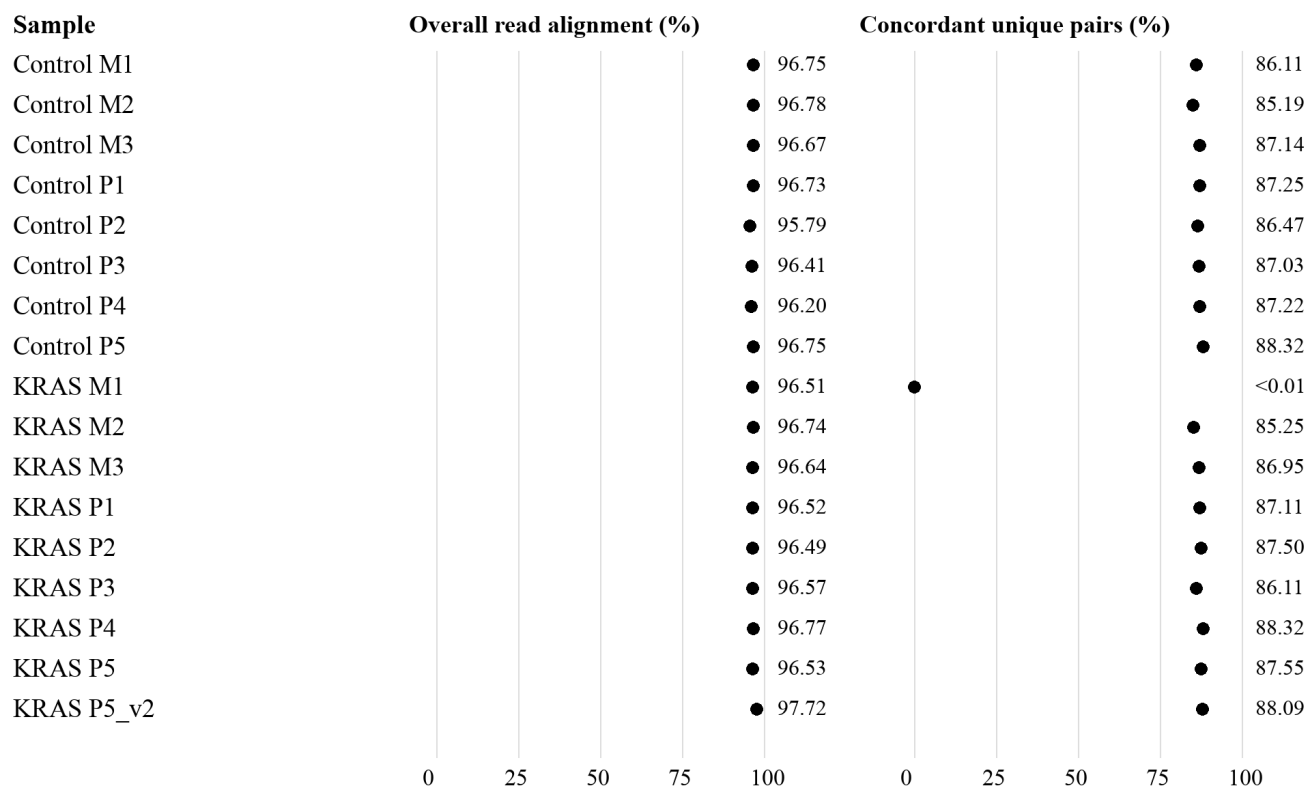
RSeQC Infer Experiment sampled 200,000 reads with a minimum mapping quality of 30 against an hg38 BED12 gene model. The two orientation fractions were 46.90%/46.73% for Control M1 and 45.80%/45.69% for KRAS P1, supporting unstranded libraries. This setting was used for the remaining runs, which were not individually tested. Across 727,731,715 input pairs, overall alignment ranged from 95.79% to 97.72% (Figure 1B). These metrics do not establish treatment effects; gene counting and differential expression were outside this report.

## Quality and alignment overview

### A FastQC module results across 34 files



### B Full run alignment rates across 17 runs



Source: verified FastQC reports and full HISAT2 summaries, 22 September 2026.

Figure 1. A: FastQC module status counts, with 34 files per module. B: overall aligned reads and concordantly uniquely aligned pairs use different denominators. All 17 full runs are shown; pilot runs are excluded.

## Appendix A Full alignment results

| Sample     | Input pairs | Overall (%) | Unique (%) | Discordant (%) |
|------------|-------------|-------------|------------|----------------|
| Control M1 | 40,237,116  | 96.75       | 86.11      | 0.61           |
| Control M2 | 45,764,716  | 96.78       | 85.19      | 0.58           |
| Control M3 | 38,181,985  | 96.67       | 87.14      | 0.64           |
| Control P1 | 45,946,458  | 96.73       | 87.25      | 0.57           |
| Control P2 | 39,989,452  | 95.79       | 86.47      | 1.15           |
| Control P3 | 47,840,900  | 96.41       | 87.03      | 0.54           |
| Control P4 | 46,287,075  | 96.20       | 87.22      | 0.65           |
| Control P5 | 45,023,981  | 96.75       | 88.32      | 0.57           |
| KRAS M1    | 45,292,678  | 96.51       | 0.0030     | 75.56          |
| KRAS M2    | 44,032,049  | 96.74       | 85.25      | 0.62           |
| KRAS M3    | 42,961,283  | 96.64       | 86.95      | 0.60           |
| KRAS P1    | 39,963,282  | 96.52       | 87.11      | 0.62           |
| KRAS P2    | 42,632,214  | 96.49       | 87.50      | 0.69           |
| KRAS P3    | 42,073,244  | 96.57       | 86.11      | 0.56           |
| KRAS P4    | 40,452,536  | 96.77       | 88.32      | 0.52           |
| KRAS P5    | 37,886,040  | 96.53       | 87.55      | 0.64           |
| KRAS P5_v2 | 43,166,706  | 97.72       | 88.09      | 0.32           |

### Calculation and interpretation

Overall (%) =  $100 \times [1 - \text{unmapped individual reads} / (2 \times \text{input pairs})]$ . Unique (%) =  $100 \times \text{concordantly uniquely aligned pairs} / \text{input pairs}$ . Discordant (%) =  $100 \times \text{discordantly aligned pairs} / \text{input pairs}$ .

For example, Control M1 overall alignment =  $100 \times [1 - 2,617,722 / (2 \times 40,237,116)] = 96.75\%$ . Its concordant unique alignment =  $100 \times 34,649,263 / 40,237,116 = 86.11\%$ .

Total input: 727,731,715 pairs. KRAS P5 is SRR24828471; P5\_v2 is SRR24828472. These are separate runs, not automatically independent biological replicates. Overall alignment alone does not establish correct pairing.

## Appendix B FastQC results

| Module                       | Files | Pass | Warn | Fail |
|------------------------------|-------|------|------|------|
| Per base sequence quality    | 34    | 34   | 0    | 0    |
| Per tile sequence quality    | 34    | 34   | 0    | 0    |
| Per sequence quality scores  | 34    | 34   | 0    | 0    |
| Per sequence GC content      | 34    | 32   | 2    | 0    |
| Per base N content           | 34    | 34   | 0    | 0    |
| Sequence Length Distribution | 34    | 34   | 0    | 0    |
| Overrepresented sequences    | 34    | 34   | 0    | 0    |
| Adapter Content              | 34    | 34   | 0    | 0    |
| Per base sequence content    | 34    | 0    | 0    | 34   |
| Sequence Duplication Levels  | 34    | 0    | 0    | 34   |

Counts refer to 34 distinct input files. Status percentage =  $100 \times \text{status count} / 34$ ; for GC content, 32/34 passed (94.12%) and 2/34 warned (5.88%). A module Fail is a diagnostic flag, not a failed software job.

## References and analysis records

1. Galaxy FastQC, MultiQC 165, HISAT2 full-run summaries and strand pilots; 22 September 2026. Reproducible records: [github.com/yichuanlin07/pdac-kas-rnaseq](https://github.com/yichuanlin07/pdac-kas-rnaseq).
2. Week 2 course slides. Quality Control and Alignment.
3. FastQC documentation. [bioinformatics.babraham.ac.uk/projects/fastqc/](https://bioinformatics.babraham.ac.uk/projects/fastqc/)
4. HISAT2 manual. [daehwankimlab.github.io/hisat2/manual/](https://daehwankimlab.github.io/hisat2/manual/)
5. RSeQC documentation. [rseqc.sourceforge.net/](https://rseqc.sourceforge.net/)
6. Klomp JA et al. Defining the KRAS- and ERK-dependent transcriptome in KRAS-mutant cancers. Science. 2024. doi:10.1126/science.adk0775.

## Author contributions

Yichuan Lin organized the data and coordinated the analysis and report preparation.

Review signature \_\_\_\_\_ Date \_\_\_\_\_